Lecture 4: 1-D spins and the Ising Model

In the calculations described so far, no interactions occur between the particles. One of the simplest examples in which interaction is allowed is the 1-D Ising spin model. The model can be used to predict macroscopic magnetic properties of solids made up of ions with a certain electron spin. In this model, interacting spins are aligned in a chain (Figure 5). A spin may flip because of its interaction with surrounding atoms and consequently neighbouring spins are affected. These spins in turn affect their neighbours, and thus one flip may propagate along the chain until all spins come to equilibrium with their surroundings.

Figure 5. Representation of electron spins as either 'up' or 'down'.

The electrons interact with one another by the exchange interaction of magnitude J. The number of electrons in each spin state, either 'up' or 'down', depends on the ratio of this interaction to the temperature. Figure 5 shows, schematically, a small section of the spin state. Parallel spin pairs have energy $-J$ but $+J$ if they are anti-parallel. If all spins are parallel, the spin system is ferromagnetic, and if anti-parallel anti-ferromagnetic.

The energy of a chain in spin state α and in the absence of an external magnetic field is given by

$$E_\alpha = -J\sum_k s_k s_{k+1}\,, \tag{12}$$

where $s_k = \pm 1$ represents the spin on the kth site, and the sum is taken over all pairs $s_k s_{k+1}$. The various combinations the spins take produce the different energies E_α. Because the exchange interaction depends on the overlap of wavefunctions, as does the related Coulomb interaction, it is of very short range, therefore, only the nearest neighbours are important, and the summation of equation (12) is pair-wise.

1-D Ising Spin Partition Function and Thermodynamic Quantities

Because the interaction couples adjacent spins, when Q_N is calculated from equation (2), the sums

become mixed producing a product of summations:

$$Q_N = \sum_{s_1=-1}^{1}\sum_{s_2=-1}^{1}\cdots\sum_{s_N=-1}^{1}\exp\left(+\frac{J}{kT}\sum_k s_k s_{k+1}\right)$$
$$= \sum_{s_1=-1}^{1}\sum_{s_2=-1}^{1}\cdots\sum_{s_N=-1}^{1} e^{Js_1s_2/k_BT}e^{Js_2s_3/k_BT}\cdots e^{Js_Ns_{N+1}/kT}$$

This expression has N terms but the last spin has index $k+1=N+1$. To deal with this a boundary

condition is necessary and we use $N+1\to 1$, thus the summation becomes,

$$Q_N = \sum_{s_1=-1}^{1}\sum_{s_2=-1}^{1}\cdots\sum_{s_N=-1}^{1} e^{Js_1s_2/kT}e^{Js_2s_3/kT}\cdots e^{Js_Ns_1/kT}$$

The result becomes apparent if the sums are expanded out first with just a few terms, and then converted to hyperbolic form. The general pattern can be seen most easily with an even number of summation terms. The result by induction to N terms is

$$Q_N = \left[2\cosh\left(J/kT\right)\right]^N + \left[2\sinh\left(J/kT\right)\right]^N .$$

In the limit of large N and except when $T\to 0$, the sinh term is small compared to the cosh, hence, to a very good approximation,

$$Q_N = \left[2\cosh\left(J/kT\right)\right]^N . \tag{13}$$

In the absence of a magnetic field, the energy, heat capacity and *entropy per spin* calculated by using equation (13) in equation (3), in equation (11), and the expression for the entropy (right after equation 3) are,

$$U = -J\tanh\left(\frac{J}{kT}\right), \tag{14}$$

$$C_V = k\left[\frac{2J}{kT}\left[\mathrm{e}^{J/kT}+e^{-J/kT}\right]^{-1}\right]^2 = k\left[\frac{J}{kT}\operatorname{sech}\left(\frac{J}{kT}\right)\right]^2, \tag{15}$$

and

$$S = k\ln\left(e^{2J/kT}+1\right) - \frac{2J}{T}\frac{e^{2J/kT}}{e^{2J/kT}+1}\,. \tag{16}$$

It is worth discussing in general terms how $\langle E\rangle$, C_V and S are expected to vary with temperature as this may provide some guide to your own calculations. Since the spins interact with one another, at very low temperature they will all align in one direction with a total energy of $-NJ$ (assuming $J > 0$) because there is not enough thermal energy to disrupt spin alignment. As the temperature increases, spins will flip over. Eventually, at infinite temperature, which in practice may be well below room temperature, the energy approaches zero because there are as many parallel as anti-parallel spin pairs.

The spin heat capacity at low temperatures is also close to zero, because there is not enough energy to flip a spin, and therefore, the rate of change of energy with temperature is zero; $C_V\left(T\to 0\right) = \left(\frac{\partial U}{\partial T}\right)_V \to 0$. At infinite temperature, the heat capacity again approaches zero. Because there are now equal populations of parallel and anti-parallel spin pairs, no more energy can be absorbed and thus the rate of change of energy with temperature is zero. As C_V is zero at both extremes of temperature and is, by definition, a positive quantity, it must pass through a maximum at some finite temperature whose value will depend on J.

A Metropolis Calculation for 1-D Ising Spins using Maple

In implementing this particular Monte Carlo calculation, there are three considerations. The first is to calculate the energy change caused by flipping one spin, the second is to decide what happens at the ends of the chain, and the third is to determine what values the spins should have at the start of the calculation.

First, to determine the energy change, a site is chosen at random and its spin flipped. The change in energy this causes is found by looking at the site and its two neighbours and calculating what happens; Figure 6 shows two examples. The change in energy $\Delta E = E_2 - E_1$ is either 0 or $\pm 4J$, and because a negative or zero change in energy is always accepted, the exponential expression in the Metropolis sampling step is always $e^{-4J/kT}$. This can be calculated outside the Monte Carlo loop, which saves considerable time.

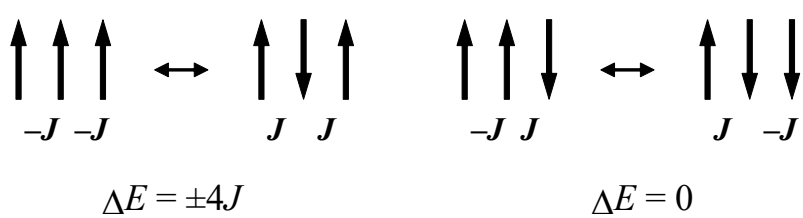

Figure 6. A schematic of two configurations of spins related by a single spin flip. Only values of

$\Delta E = 0,\ \pm 4J$ occur. The interaction energy is $-J$ between a pair of parallel spins whether up or down, and $+J$ if anti-parallel. The energy change is calculated as final – initial state.

The second consideration concerns the finite length of the chain. Theoretically, infinite length is assumed, but obviously this must be finite and thus a boundary condition is necessary to simulate infinity. The boundary condition used effectively makes the chain circular, so that if a spin's nearest neighbour is one step off the end of the chain, the state of the spin at the other end is used. Therefore, if the indices run from 1 to N then index $N+1 \rightarrow 1$ and index $0 \rightarrow N$. These steps are labelled in the Maple code shown below. The last consideration concerns the initial arrangement of the spins. Are these to be aligned, to be random, or somewhere in between? Although it does not matter how the initial spins are arranged provided enough Monte-Carlo runs are taken, at low temperatures fewer runs are needed if the spins are initially parallel.

Problem 4: Workshop on 1-D Ising spins

Pseudo Code

```
Seed = randomize()              # start random number generator
N = 500                         # N sites
L = N*2000                           # 2000 calcs / site
J = 1.0                         # positive J is ferromagnet
k = 1.0                         # Boltzmann const, in a.u.
T = 2.0                         # temperature
rnd =   rand(1..N)              # random number in range 1 to N
spin =  Array(1..N)
# deltaE can take values ( -4, 0, 4), however, only the
# positive value is used in the metropolis test, hence we define
# probability P for the Metropolis test outside the loop.
P= exp(-4.0*J/(k*T))                 # prob'lty in Metropolis test.
for i from 1 to N do
    spin[i] = -1                     # initial distribution
end do
Enrg = -J*N                          # all spins aligned initially
tot_Enrg = 0
tot_Enrg2 = 0
```

```
for num from 1 to L do                # start Metropolis
    i = rand()                        # choose spin at random
                                      # Boundary condition follow
    indx1 = i+1
    if indx1 > N then indx1= 1 end if
    indx2 = i-1
    if indx2 < 1 then indx2 = N end if
    DeltaE = 2*J*spin[i]*(spin[indx1] + spin[indx2])
                                      # Metropolis test next
    if DeltaE <= 0 or rand() < P then # rand between 0 and 1
        spin[i] = -spin[i]                 # flip spin
        Enrg = Enrg + DeltaE               # add in energy
    end if
    tot_Enrg = tot_Enrg + Enrg
    tot_Enrg2 = tot_Enrg2 + Enrg^2
end do                                # end metropolis loop
temperature = T
avE =  tot_Enrg/L                     # total <E > , N sites
avE2 = tot_Enrg2/L                    # total <E^2> , N sites
`<Energy>/spin`= avE/N
`<Energy squared>/spin`= avE2/N
Cv = 1/(k*T^2)*(avE2 - avE^2)/N
```

In the calculation shown in Figure 7 below, a chain of 500 spins is simulated and the number of spin states is therefore $2^{500} \approx 3\times10^{150}$, all of which would have to be evaluated in a direct calculation. The results of the Monte Carlo calculation, Figure 7, have been obtained by randomly sampling each site 4000 times. While this represents a large number of calculations, it is only a miniscule fraction of the total number of possible spin states and demonstrates the power of this algorithm. The solid lines, Figure 7, are calculated with the analytical solution to the partition function whose derivation is also outlined above. It is clear that the Metropolis

algorithm accurately simulates the change in energy and heat capacity with temperature.

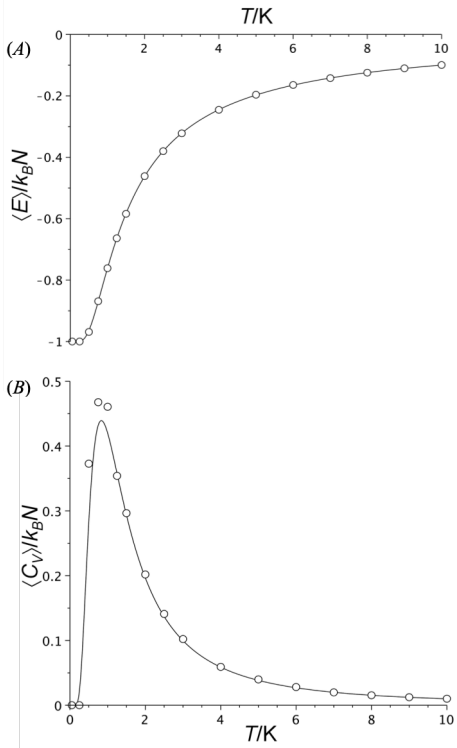

Figure 7. (*A*) The open circles are the Monte-Carlo calculated energy per spin $\langle E\rangle/kN$ and (*B*) the heat capacity per spin $\langle C_V\rangle/kN$ vs. temperature (K), with $J/k = 1$ K. One run of 4000 samples is shown for each point. The average of five runs of $\langle C_V\rangle/kN$ at $T = 1$ K has a mean of 0.45 and a 95% confidence limit of ±0.015. The lines are from the analytical functions given in equations (14) and (15).

Task 1: **(3 marks)** Perform Monte Carlo simulations to make the corresponding plot for the heat capacity, i.e. plot $\langle C_V\rangle/kN$ vs. temperature with $J/k = 0.5$ K, and compare to corresponding analytical solution. What is the effect of decreasing the coupling energy and why?

Task 2: **(2 marks)** Calculating the Entropy

Although the entropy at a given temperature T can be calculated by integrating C_V/T, it can also be calculated from the arrangement of the spins. Maps of the spin state at the end of the calculation indicate how the entropy changes with temperature, and three examples are shown in Figure 8A. Informally, we may label a black group as 'parallel spin up' and a grey one 'parallel spin down', although 'up' and 'down' should not be taken literally. Within any blocks of either colour the spins are parallel to one another, hence the coupling is $-J$ between adjacent spins. A colour boundary thus indicates spin pairing and a coupling of $+J$ between these two spins; see Figure 8. At $T = 0$ K all spins are in the same state, which would mean that the whole strip would be of the same colour. However, there are two ground states at $T = 0$ and these could be coloured

either black or grey. At low temperatures, e.g. T = 1/2 or 1 K, (Figure 8A) spins are grouped into a few large blocks of similar spin state. Although only one example is given in the figure, it is clear that if many calculations are performed at, say, T = 1/2 K, only a limited number of arrangements of these large blocks of spins will be possible. These configurations or arrangements, determine the entropy. As the temperature is increased, the coupling between spins becomes relatively less important compared to the energy supplied by the surrounding heat bath. This reduces the size of the groups of similar spin state but increases their number, which leads to more of ways of arranging them, and hence, a larger entropy.

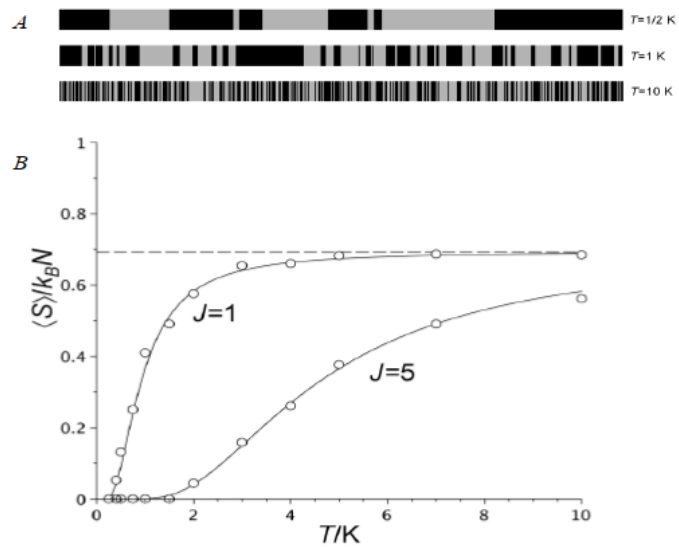

Figure 8. (A) Examples of spin states for 500 spins and 2000 samples / spin and with J/k = 1 K at T = 1/2, 1 and 10 K. The pattern at T = 1/2 K contains 11 groups since the black areas correspond to one spin state and grey the other. (B) The open circles show the Monte Carlo calculated spin entropy as $\langle S \rangle / kN$ vs. temperature with J/k = 1 K , J/k = 5 K and using equation (17). At each temperature and J value N = 500 and 2000 samples/spin were taken. The solid lines are calculated with the theoretical function equation (16).

Your task is to:

a) Calculate the entropy S for a given J value at five different temperatures.

b) Plot the results in a graph as shown in Figure 8B (including theoretical and Monte Carlo data).

c) Explain the results for the entropy in the limits when T -> zero and T -> infinity.

Hint: The entropy of the equilibrium state is given by $S = k\ln(\Omega)$, where Ω is the number of arrangements that the spins can take. To calculate Ω, the number of groups of spins p present at the end of the calculation and at a given temperature needs to be counted (see Figure 8A). Your

task is to implement this in the code.

From the theory of statistics, the number of ways of choosing p distinguishable objects from a group of N distinguishable objects, where the order of choosing does not matter is given by

$$\Omega' = \frac{N!}{p!(N-p)!} \tag{17}$$

This is the number of arrangements, or combinations. Equation (17) is a good approximation to the number of arrangements of groups of spins, when N is large and the temperature is not zero, thus $\Omega \approx \Omega'$ .The good agreement, Figure 8(*B*), between the equation (16) describing the entropy per spin and that obtained using equation (17) shows that this is a good approximation. Within statistical variation, the entropy per spin obtained by counting groups of spins and using equation (17), and that calculated from theory are the same, see Figure 8(*B*).

Task 3: **(2 marks)** Einstein model for solids

(a) Derive/calculate theoretically expressions for U, C_V, and S in the limit of $J = 0$ and in the presence of a magnetic field B and depict their T dependence assuming convenient values for the parameter.

(b) Perform Monte Carlo simulations to calculate U, C_V, and S in the limit of $J = 0$ and in the presence of a magnetic field B and depict their T dependence.

Task 4: **Quality** of presentation of report in Jupyter notebook **(2 marks)**

Task 5: **Bonus point (1 mark).** Justify why your work/report goes beyond just completing the tasks.

Submission of work:

- The work must be submitted as an ipynb on minerva
- The filename must be Surname-problem4.ipynb
- The self-assessment must be added at the end of the ipynb

Self-assessment:

- Aim is that you give an honest reflection on how you completed the tasks
- You will need justify all your marks
- Declaration of integrity (That the work you submitted is yours, and that you marked it fairly)

Deadline: 2nd March 2026 at 5pm.

Final remarks

Equivalence of ensembles Finally, we note that $S = k\ln(\Omega)$ is derived using a microcanonical ensemble, (N, V, E constant) whereas the entropy calculated using the partition function is derived from a canonical ensemble (N, V, T constant). Energy fluctuations are permitted in the canonical ensemble but energy is fixed in the microcanonical ensemble. This inherent difference does not contradict their equivalency, provided that the size of the fluctuations becomes vanishingly small in the canonical ensemble, which it does in the limit of large N. The argument to demonstrate this follows along the lines of that of equation (9) where the energy dispersion σ (equation 8) is now interpreted as the energy fluctuation of the canonical ensemble. The ratio of energy fluctuation to average energy $\sigma / \langle E \rangle \approx 1/\sqrt{N}$, becomes vanishingly small as N increases.

Phase Change The gradual and continuous change in the heat capacity and entropy indicates that the phase transition between ordered and disordered spins is unlike the familiar first-order or 'all-or-none' phase transitions or even that in the 2-D Ising spin model. While the general change in C_V and S with temperature is similar to that of a normal substance (O_2, H_2O etc.), the detail is quite different. This is because in the 1-D spin case the heat capacity increases and decreases in a continuous manner with increasing temperature. Similarly, the entropy smoothly increases with temperature to reach its final value. In a normal substance, however, there is a step change both in C_V and S at a phase boundary. The cause of the different behaviour is because, energetically, it is more favourable for two regions of opposite spins to co-exist than it is for either one or the other to exist on its own.